

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [219, 153, 156, 122, 119, 143, 130, 110, 120, 126]

Tap amplitudes (pixels): [120.7683733433551, 128.03515142334936, 125.39936203984452, 108.24047302187847, 93.77632963600144, 89.47066558375433, 100.4987562112089, 90.4267659490264, 92.17917335276988, 90.78546139112804, 110.57124400132251]

GPT-5

Analysis of extracted features for MDS-UPDRS 3.4 (Finger Tapping)

- Task target: 10 taps; extracted shows Number of taps = 11 with 10 tap intervals provided. This suggests 11 peaks detected spanning 10 inter-tap periods, which is consistent for 10 consecutive taps.

Speed (periods)

- Periods (frames): [219, 153, 156, 122, 119, 143, 130, 110, 120, 126]
- Trend:
 - Clear initial speeding-up from 219 → 153–156 → 122–119 (faster taps as periods drop).
 - Then fluctuation: 143 (slower), 130 (faster), 110 (fastest), followed by 120–126 (slight slowing).
- Interpretation:
 - No persistent moderate or severe slowing. The variability shows a small mid-sequence slowing (period 143) and end-sequence mild slowing (120 → 126), but overall speed is fairly brisk after the first few taps.
 - No evidence of prolonged arrests/freezes.

Rhythm interruptions/hesitations

- Using period variability as a proxy, we see a few irregularities (notably at period 143 after sustained faster periods). This may reflect intermittent hesitation.

- Estimated interruptions/hesitations: likely 1–2 brief irregularities, possibly up to 3 depending on thresholding. There is no sign of >5 interruptions or any clear prolonged arrest.

Amplitude and decrement

- Amplitudes (pixels): [120.8, 128.0, 125.4, 108.2, 93.8, 89.5, 100.5, 90.4, 92.2, 90.8, 110.6]
- Pattern:
 - Starts high (first 2–3 taps), then a noticeable drop by tap 4–6 (108 → 93.8 → 89.5).
 - Partial recovery mid-to-late (100.5) but remains lower than start for most of the sequence (90–100 range), with a late rebound at the final detected tap (110.6).
- Interpretation:
 - There is a clear decrement in amplitude that begins before the midpoint (from tap 4 onward), i.e., starting after the 1st tap and becoming evident by taps 4–6. The overall pattern shows sustained reduction compared to the initial taps, aside from a late rebound.
 - This is more than just a slight end-of-sequence decrement.

Mapping to MDS-UPDRS criteria

- Slight (1): slight slowing or end-only decrement or 1–2 hesitations. The amplitude decrement here is not restricted to near the end; it emerges early/mid and persists.
- Mild (2): 3–5 interruptions OR mild slowing OR amplitude decrement midway through the 10 taps. The amplitude decrement becomes evident by around tap 4–6 (midway), matching “amplitude decrements midway.”
- Moderate (3): >5 interruptions or at least one longer arrest OR moderate slowing OR amplitude decrement starting after the 1st tap. Although the decrement does start after the first tap, the overall clinical impression from speed and rhythm does not reach “moderate” severity: no freezes, no >5 interruptions, and speed is generally adequate with only mild variability. The amplitude reduction is notable but partially recovers and is not consistently severe throughout.

Given the balance of features—mid-sequence amplitude decrement with mild speed variability and at most a few brief hesitations—the best fit is: